

Programming and Systems Development

Final Lab Exam – Mock Unix - **Total Marks: 40**

Unix Exam Mock paper (there is no past paper because it is the first time we have the Unix Exam. **Note: A typical exam paper will also include some examples of expected outputs**)

You should create a single bash script file and write the commands for each task in the same bash script file. You should name the file `your_name_unix_task.sh` (fill in your own name). It is expected that once the bash script file is run, it will automatically perform all the tasks.

Orientation & File Processing (Total marks:14)

You are provided with a file named **server_access.log** containing connection records.

Important

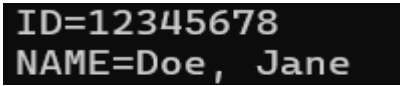
- Do *all* work in a directory called **unix_practice** under your home directory.
 - Use the exact filenames and formats specified.
 - Only use `sudo` where explicitly stated.
 - Some tasks want **only command outputs**, others want **command + output** (read carefully).
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Task 1a: Navigation and Basic Redirection

1. Download and copy the file **server_access.log** to your home directory in the Linux remote server: **10.224.160.71**. **[1 mark]**
2. Create a directory named **unix_practice**, move the log file into it, and inside create a file called **student_record.txt** containing exactly three lines:

Name: YourName

ID: YourStudentID

Example: 

[2 marks]

3. Change directory into `~/unix_practice`.
[1 mark]

4. Display all files including hidden ones and redirect the listing to a file named **dir_overview.txt**.
[1 mark]
5. Show lines 11–20 (inclusive) of `server_access.log` using any appropriate command and redirect to **segment.txt**.
[1 mark]

Task 1b: Log Filtering and Counting

1. Extract all lines containing the word **FAILED** (case-insensitive) into **failed_attempts.txt**.
[2 marks]
2. From that file, extract only usernames (use text tools or regex, assuming the username appears after `user=`) and save to **usernames.txt**.
[3 marks]
3. Count how many times each username appears, sorted from least to most frequent. Save both the command and its output to **fail_summary.txt**.
[3 marks]

Section 2: Users, Permissions, and Processes (16 marks)

Task 2a: Ownership and Permissions

1. Create a text file named **maintenance.log**.
[1 mark]
2. Give the file permissions `rw-r-----`.
[1 mark]
3. Display details with `ls -l` and redirect that output to **perm_check.txt**.
[1 mark]
4. Create a group **Netops** and a user **analyst** (if they do not exist). Append the exact command lines used to create both to **perm_check.txt**.
[3 marks]
5. Change ownership of **maintenance.log** to `analyst:netops`. Then try appending a line to the file as your current user — capture the resulting message (success or error) and append to **perm_check.txt**.
[2 marks]

Task 2b: Process Monitoring

1. Launch three background jobs:
 - `ping -i 2 localhost`
 - `sleep 1500`

- yes testing
[1 mark]
- 2. List all active jobs with their PIDs and redirect to **job_monitor.txt**. [1 mark]
Identify and append the PID of the yes testing job to that same file.
[1 mark]
- 3. Suspend or terminate the yes process using a suitable command and log both the command and confirmation message into **job_monitor.txt**.
[2 marks]
- 4. Show that only the other two background jobs remain.
Include both the command and its output in **job_monitor.txt**.
[3 marks]

Section 3: Archiving & File Investigation (10 marks)

You are provided with a compressed archive called **logs_archive.zip**.

Task 3a: File Investigation

1. Extract the archive into your **unix_practice** folder.
[2 marks]
2. Inside the extracted folder, identify file types using file * and redirect output to **types_report.txt**.
[2 marks]
3. Search recursively within the extracted folder for any occurrence of the string **root@** and redirect all matching lines (with filenames) into **root_refs.txt**.
[2 marks]
4. Display the first 16 bytes (in hex) of any file whose name includes "auth" and redirect both the command and its output to **auth_hex.txt**.
[2 marks]

Task 3b: Compression

1. Create a compressed tarball of your **unix_practice** directory named **studentID_exam_logs.tar.gz** (replace with your own ID).
[2 marks]

Unix submission

Ensure your submission file is named exactly as it is supposed to (i.e. **your_name_unix_task.sh**) and that it contains the code for all of the Unix tasks. It is expected that once the bash script file is run, it will automatically perform all the tasks.